

Anish Kumar

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PROFILE

B.Tech Information Technology student at IIIT Bhopal focused on AI Infrastructure, ML Systems, Distributed Systems and High-Performance Computing. Building strong foundations in C++, Python, Linux, concurrency, networking, CUDA, GPU programming, distributed training and LLM inference systems.

EDUCATION

Indian Institute of Information Technology, Bhopal

B.Tech in Information Technology | Expected May 2027

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Systems, Computer Architecture, Probability & Statistics, Linear Algebra

EXPERIENCE

VRISHTI — Front-End Developer & Graphic Designer | SIH Project

React.js, Canva

- Contributed to a contract-farming web platform using React.js and created supporting visual assets.
- Worked on responsive interfaces and user-facing components in a collaborative project environment.

SYSTEMS / AI INFRASTRUCTURE PROJECTS

vLLM — LLM Inference & Serving

Python, C++, CUDA, PyTorch

- Studied high-throughput LLM serving, continuous batching, KV-cache management, PagedAttention and GPU-aware scheduling.
- Benchmarked inference workloads and analyzed latency, throughput, GPU utilization and memory bottlenecks.

GitHub: github.com/vllm-project/vllm 

DeepSpeed — Distributed Training Infrastructure

Python, C++, CUDA, PyTorch

- Studied distributed training, data/model parallelism and memory-efficient large-model training workflows.
- Analyzed multi-GPU compute, memory and communication bottlenecks using profiling-oriented workflows.

GitHub: github.com/deepspeedai/DeepSpeed 

Triton — GPU Kernel & Compiler Systems

Python, C++, MLIR, LLVM

- Studied custom GPU kernels, tiling, memory access, parallel execution and kernel-level performance optimization.
- Explored Triton compiler concepts, MLIR/LLVM and GPU code generation.

GitHub: github.com/triton-lang/triton 

TECHNICAL SKILLS

- **Programming:** C++, Python, Bash, SQL, Git
- **Systems:** Linux, Processes, Threads, Multithreading, Synchronization, Concurrency, TCP/IP, HTTP, gRPC, Protocol Buffers
- **Backend / Data:** FastAPI, REST APIs, PostgreSQL, Redis, Kafka
- **Infrastructure:** Docker, Kubernetes, Helm, GitHub Actions, Prometheus, Grafana, OpenTelemetry
- **ML Systems:** PyTorch, TensorFlow, CUDA, Triton, vLLM, DeepSpeed, Distributed Training, LLM Inference
- **Performance:** GDB, LLDB, Valgrind, AddressSanitizer, ThreadSanitizer, perf, Nsight Systems, Google Benchmark